

Expt 7	Spectral Estimation
Date:	7.2 Analysis of Windowing Effects on Spectral Estimation Using Rectangular, Hamming, and Hanning Windows in MATLAB

CODE :-

```
clc; clear; close all;
```

```
% Signal
```

```
fs = 1000; N = 256; t = (0:N-1)/fs;
```

```
x = sin(2*pi*100*t) + 0.8*sin(2*pi*110*t);
```

```
NFFT = 2048; f = (0:NFFT/2-1)*(fs/NFFT);
```

```
% Windows
```

```
w = {rectwin(N), hamming(N), hann(N)};
```

```
names = {'Rectangular','Hamming','Hanning'};
```

```
% Processing
```

```
for i = 1:3
```

```
    xw = x(:).*w{i};
```

```
    X = fft(xw,NFFT);
```

```
    P{i} = abs(X(1:NFFT/2));
```

```
    P{i} = P{i}/max(P{i});
```

```
end
```

```
%% Individual Spectrums
```

```
figure;
```

```
for i = 1:3
```

```
    subplot(3,1,i);
```

```
    plot(f,20*log10(P{i}),'LineWidth',1.5);
```

```
title(['Spectrum using ' names{i} ' Window']);
```

```
xlim([0 250]); grid on;
```

```
end
```

```
%% Comparison
```

```
figure;
```

```
plot(f,20*log10(P{1}),'k',f,20*log10(P{2}),'r',f,20*log10(P{3}),'b','LineWidth',1.5);
```

```
legend(names); title('Window Comparison'); xlim([50 170]); grid on;
```

```
%% Window Shapes
```

```
figure;
```

```
for i = 1:3
```

```
subplot(3,1,i); plot(w{i},'LineWidth',1.5);
```

```
title([names{i} ' Window']); grid on;
```

```
end
```

```
%% Peak Detection
```

```
for i = 1:3
```

```
[pks{i},loc{i}] = findpeaks(P{i},f,'MinPeakHeight',0.3);
```

```
fprintf('\n%s Window Frequencies:\n',names{i});
```

```
disp(loc{i});
```

```
end
```

```
%% Peak Marking (Hamming)
```

```
figure;
```

```
plot(f,20*log10(P{2}),'LineWidth',1.5); hold on;
```

```
plot(loc{2},20*log10(pks{2}),'ro','MarkerFaceColor','r');
```

```
title('Detected Frequencies (Hamming)'); xlim([50 170]); grid on;
```

```

%% Main Peaks

fprintf('\nMain Spectral Peaks:\n');

for i = 1:3

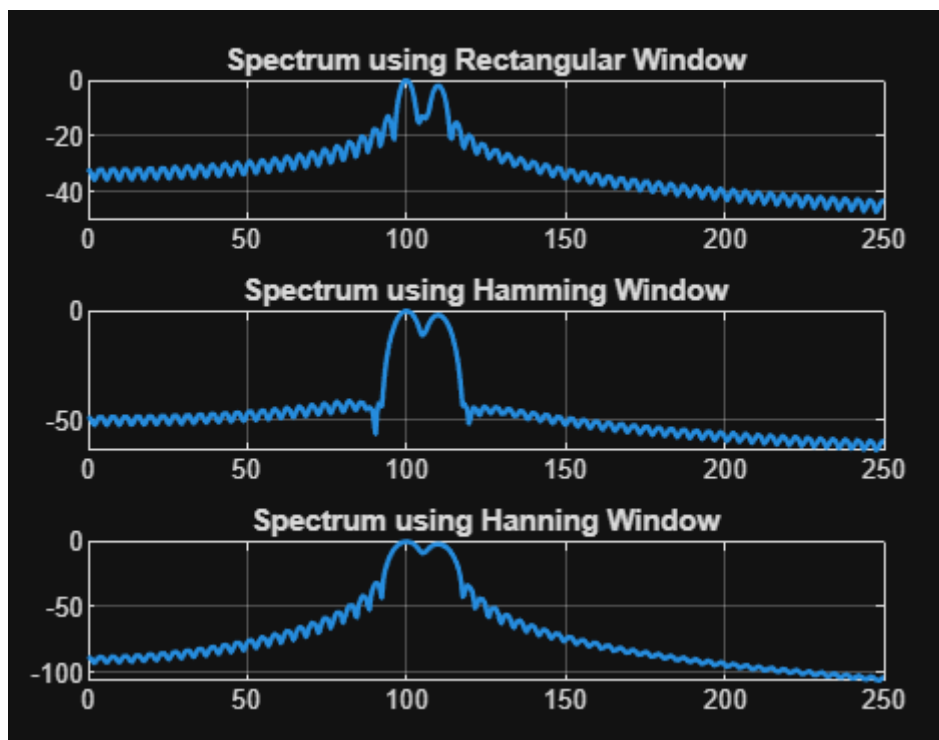
[~,idx] = max(P{i});

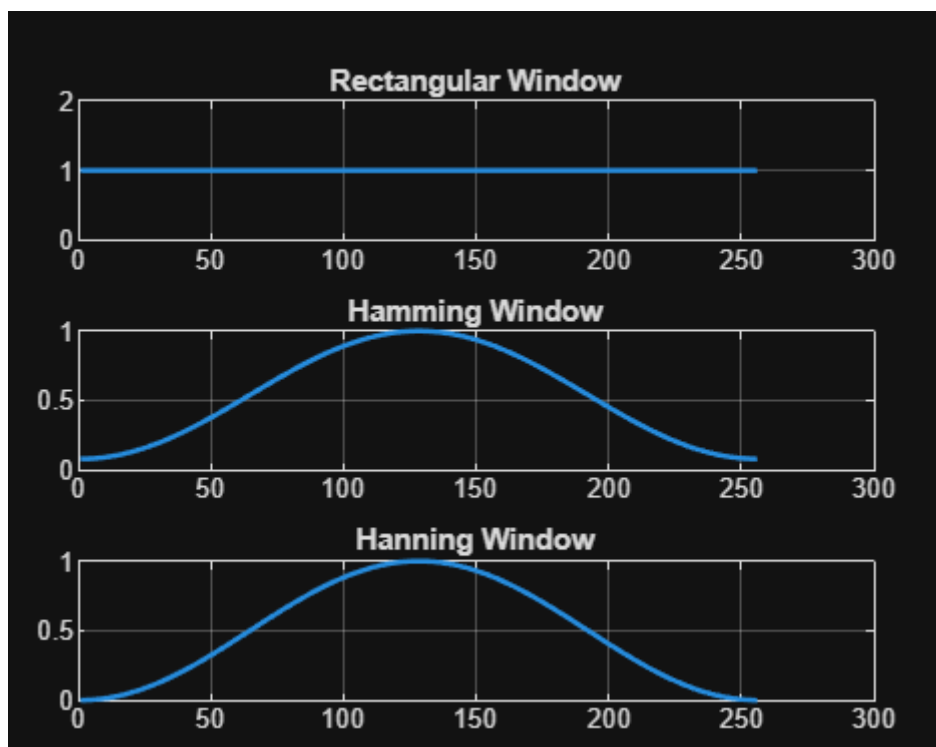
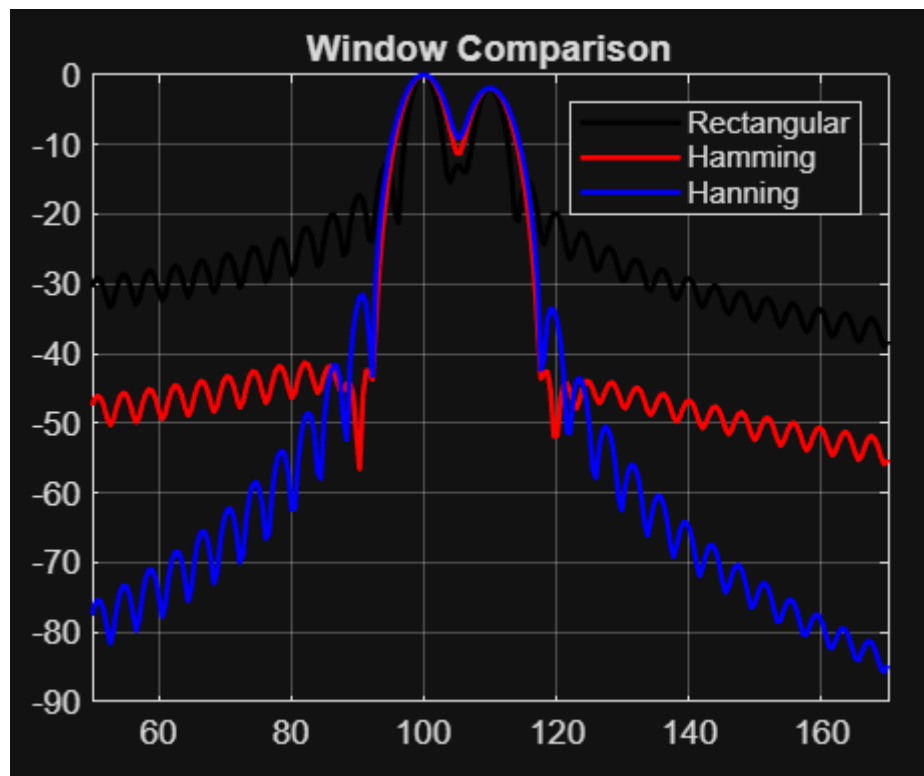
fprintf('%s Window Peak = %.2f Hz\n',names{i},f(idx));

end

```

OUTPUT :-





Rectangular Window Frequencies:

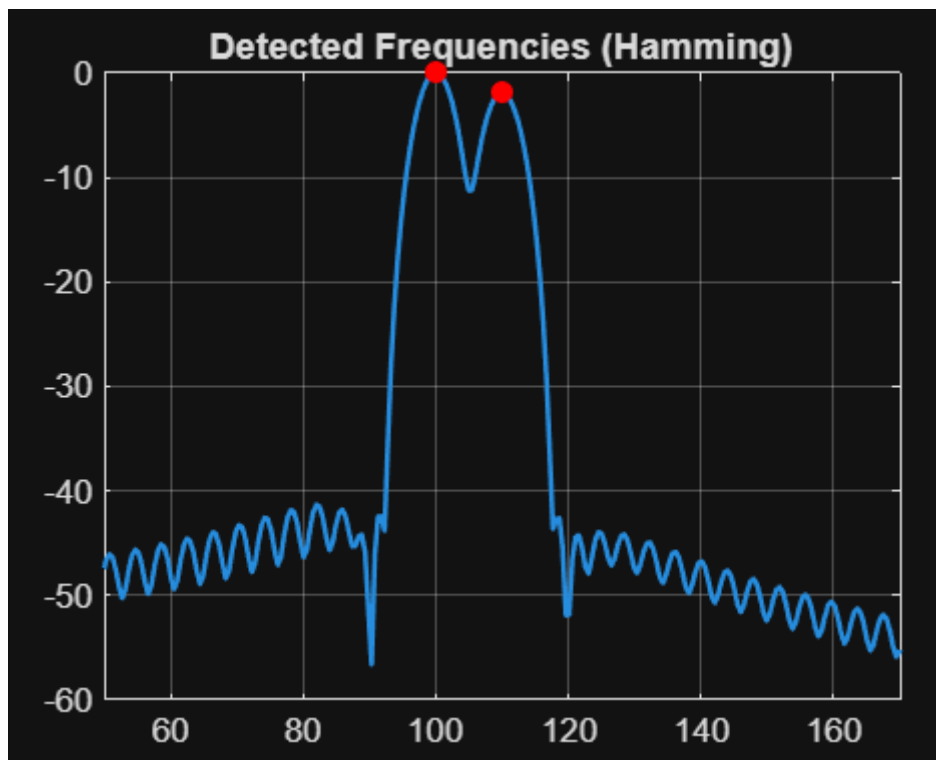
100.0977 109.8633

Hamming Window Frequencies:

100.0977 109.8633

Hanning Window Frequencies:

100.0977 109.8633



Main Spectral Peaks:

Rectangular Window Peak = 100.10 Hz

Hamming Window Peak = 100.10 Hz

Hanning Window Peak = 100.10 Hz